IIT ROORKEE

NPTEL ONLINE CERTIFICATION COURSE

Charging Infrastructure

Lecture-50

DC Charging (CCS2)-I

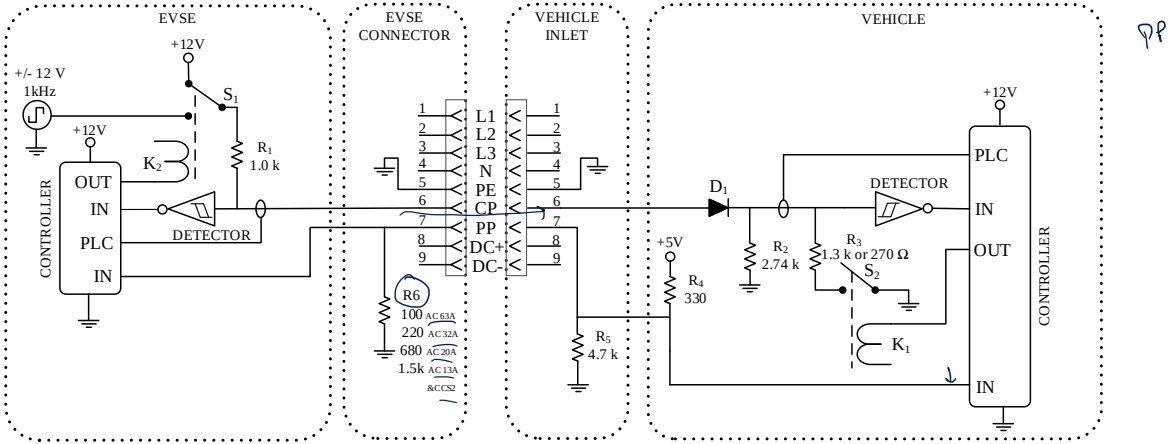
Dr. Apurv Kumar Yadav

Department of Electrical Engineering



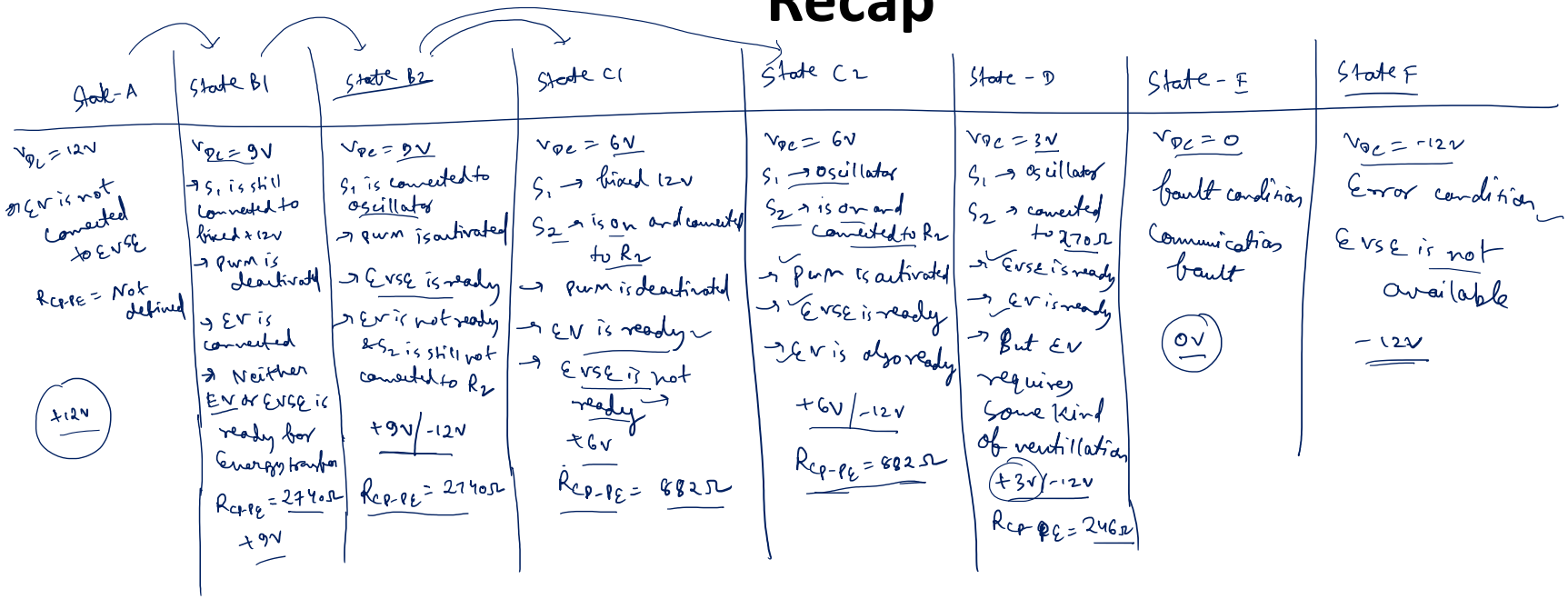
Recap

- AC Type-2 Charging : Communication, charging sequence



The details are in given in SAEJ1772 standards (in the form of SAE J1772 signalling protocol) and also adopted in IEC 61851

Recap

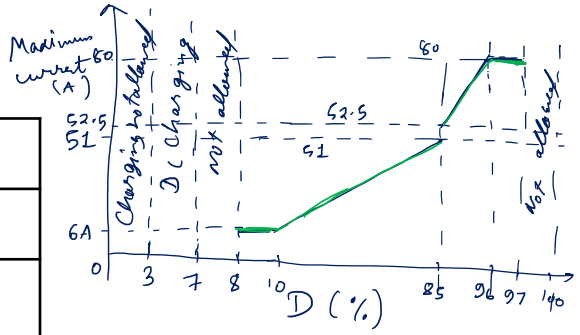


The details are in given in SAEJ1772 standards (in the form of SAE J1772 signalling protocol) and also adopted in IEC 61851

Recap

CP Signal Duty Ratio

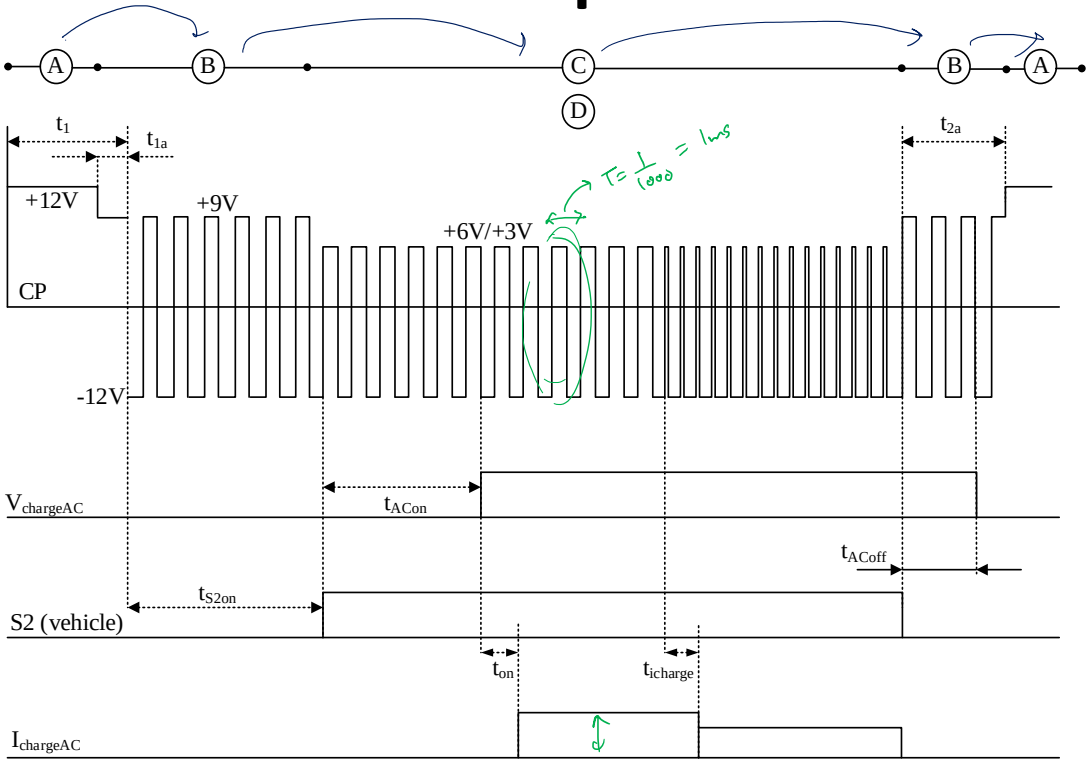
Duty ratio (%)	Maximum current (A) to be drawn by EV
0 ≤ D ≤ 3	Not allowed
3 ≤ D ≤ 7	As indicated by digital communication (high-level communication) for DC charging
7 ≤ D ≤ 8	Not allowed
8 ≤ D ≤ 10	6
10 ≤ D ≤ 85	0.6*D
85 ≤ D ≤ 96	(D - 64) * 2.5
96 ≤ D ≤ 97	80
97 ≤ D ≤ 100	Not allowed



Maximum current to be drawn by vehicle according to IEC 61851-1

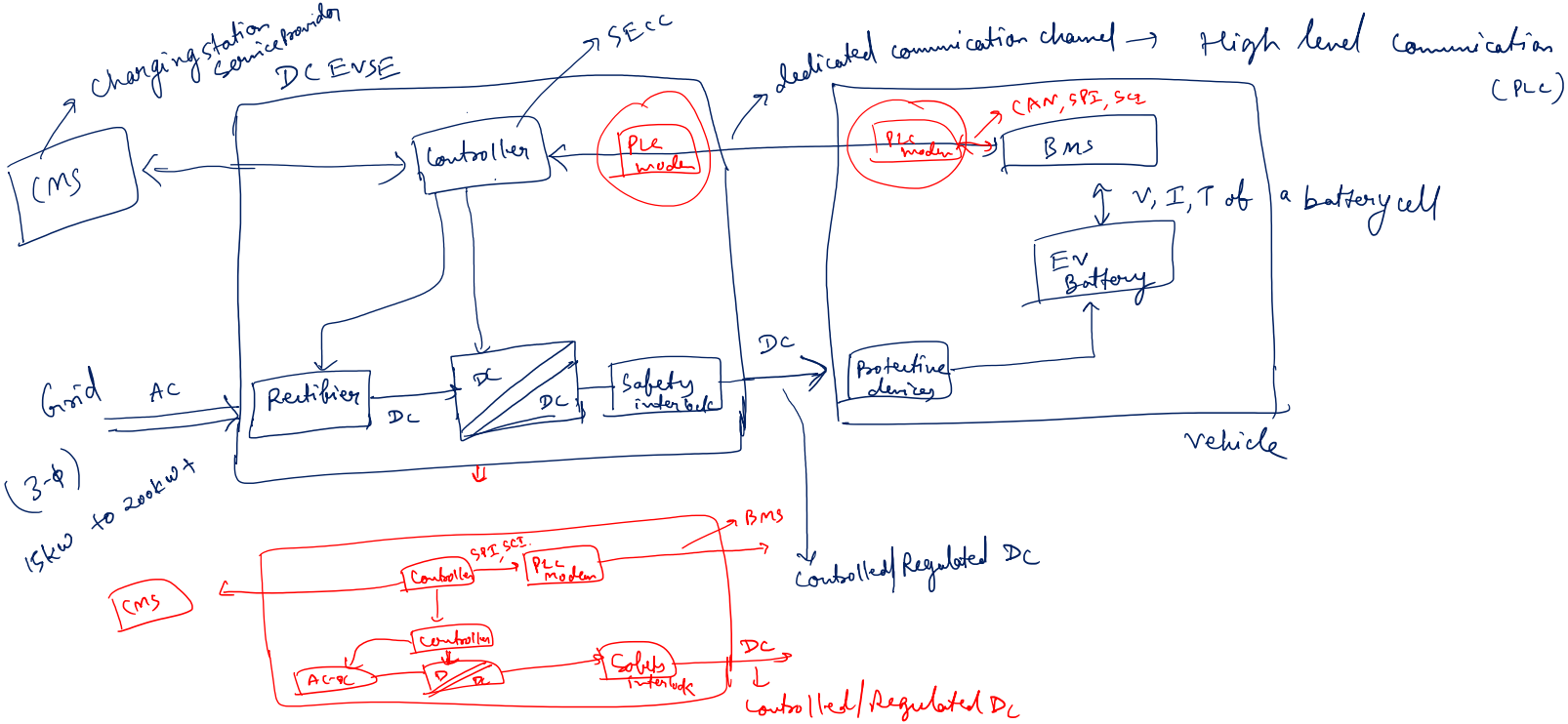
Bahrami, Ali. (2020). EV Charging Definitions, Modes, Levels, Communication Protocols and Applied Standards. 10.13140/RG.2.2.15844.53123/11.

Recap: Waveform at CP pin

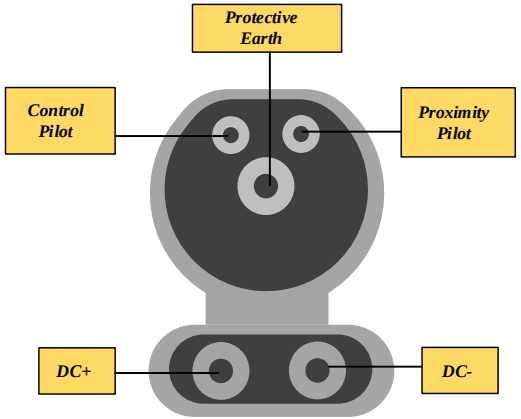


Source: M. Parchomiuk, A. Moradewicz and H. Gawiński, "An Overview of Electric Vehicles Fast Charging Infrastructure," *2019 Progress in Applied Electrical Engineering (PAEE)*, Koscielisko, Poland, 2019, pp. 1-5, doi: 10.1109/PAEE.2019.8788983

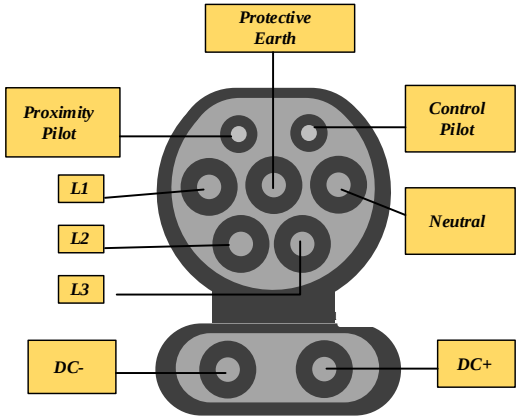
Recap: DC Charging System



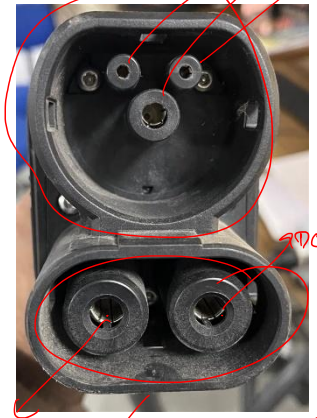
DC Charging (CCS2)



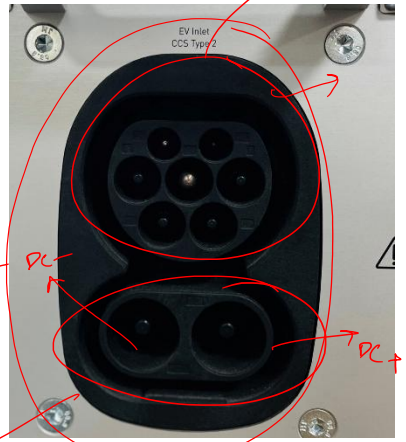
Charger Plug



Vehicle Inlet Connector

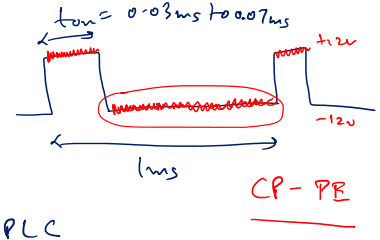


Charger Plug



Vehicle Inlet Connector

DC CCS2 Charging System

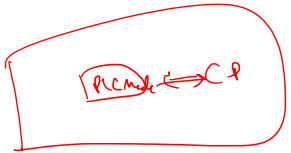


- PP circuit is same as that of the AC Type-2 connector
- The communication → indirectly over CP (Control Pilot) and PE pins.
- CP signal width is kept within 3-7 % range, in addition to this, there is high frequency modulated signals which is being superimposed on 1kHz PWM over CP and PE signals to enable power line communication (PLC).
- The required parameters and communication method are specified in ISO 15118 standard and DIN SPEC 70121.
- The structure of the communication protocol is based on the seven-layer network model called OSI (Open Systems Interconnection Reference Model).

Courtesy: 1. Rata M,et.al " The ElectricalVehicle Simulator for Charging Station in Mode 3 of IEC 61851-1 Standard. *Energies*. 2020; 13(1):176. https://doi.org/10.3390/en13010176
2. : M. Parchomiuk, et.al, "An Overview of Electric Vehicles Fast Charging Infrastructure," *2019 Progress in Applied Electrical Engineering (PAEE)*, Koscielisko, Poland, 2019, pp. 1-5, doi: 10.1109/PAEE.2019.8788983

DC CCS2 Charging System Cont.

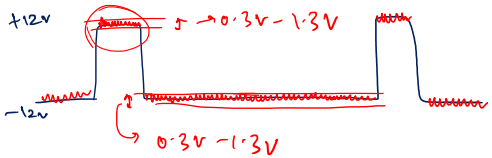
- Based on the IEC-61851 Base communication, the duty-cycle is fixed to 3 to 7%.
- The PLC signal (power line communication) is modulated onto the Control Pilot (CP) with 3 to 7% PWM signal.
- The PLC signal contains High-Level protocol messages.
- The physical interface uses HomePlug Green PHY standard along with Signal Level Attenuation Characterization (SLAC).
- The correct digital data, appropriately modulated, is applied via PLC modem connected to the CP line (having IEC 61851-1).



Courtesy: 1. Rata M,et.al " The ElectricalVehicle Simulator for Charging Station in Mode 3 of IEC 61851-1 Standard. *Energies*. 2020; 13(1):176. https://doi.org/10.3390/en13010176
2. : M. Parchomiuk, et.al, "An Overview of Electric Vehicles Fast Charging Infrastructure," *2019 Progress in Applied Electrical Engineering (PAEE)*, Koscielisko, Poland, 2019, pp. 1-5, doi: 10.1109/PAEE.2019.8788983

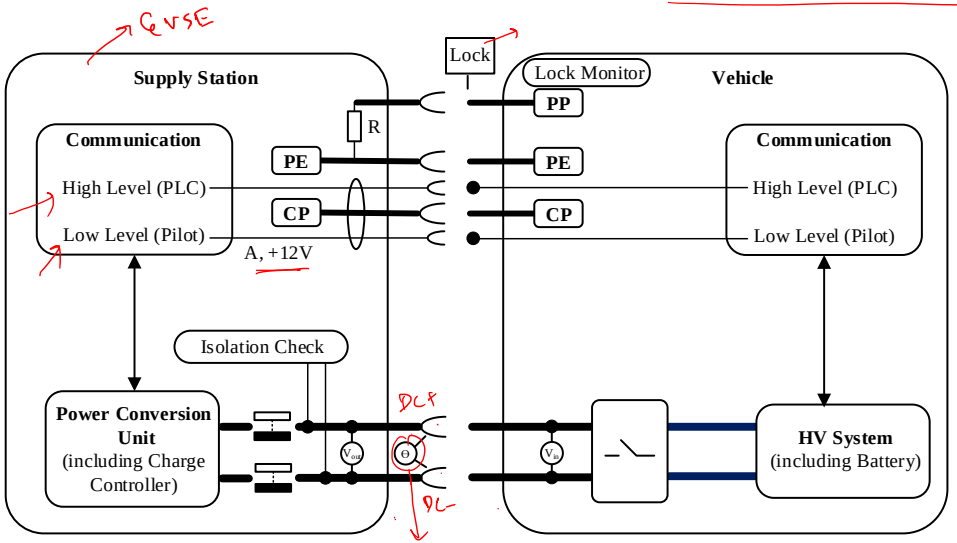
DC CCS2 Charging System Cont.

- HomePlug Green PHY (HPGP) based PLC uses OFDM amplitude-phase modulation (Orthogonal Frequency-Division Multiplexing), which separates the fast broadband transmission into many slower narrowband transmissions.
- Data are transmitted over 2–28 MHz wideband OFDM.
- Total of 1155 subcarriers with 917 of them are used.
- The subcarrier frequency signal is generated with a 24.414 kHz interval.
- The subcarrier modulation is done using QPSK.
- The HPGP digital data transmission rate is 4.5 to 10 Mb/s.
- Peak-to peak PLC signal is in the range of 0.3Vpp–1.3Vpp.



1. Ramasamy, D. Possible Hardware Architectures for Power Line Communication in Automotive V2G Applications. *J. Inst. Eng. India Ser. B* **104**, 813–819 (2023). https://doi.org/10.1007/s40031-023-00880-5.
2. Bahrami, Ali. (2020). EV Charging Definitions, Modes, Levels, Communication Protocols and Applied Standards. 10.13140/RG.2.2.15844.53123/11.

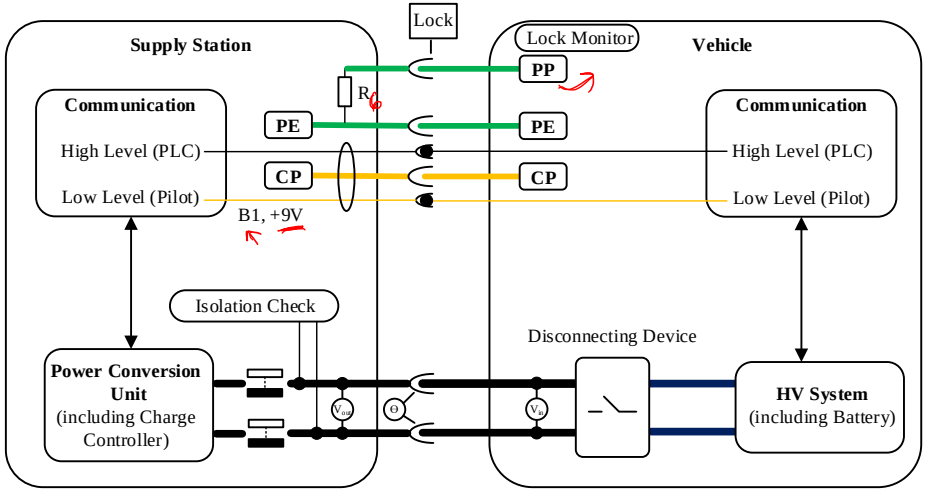
DC CCS2 Charging Sequence



- Charger plug is Unmated
- Charging is in State-A

Source: 1. https://morth.gov.in/sites/default/files/ASI/320201825106PMAIS_138_Part_2_F.pdf
2. https://www.charin.global/media/pages/technology/ccs-specification/42a9d61e04-1626949173/design_guide_combined_charging_system_v7.pdf

DC CCS2 Charging Sequence Cont.



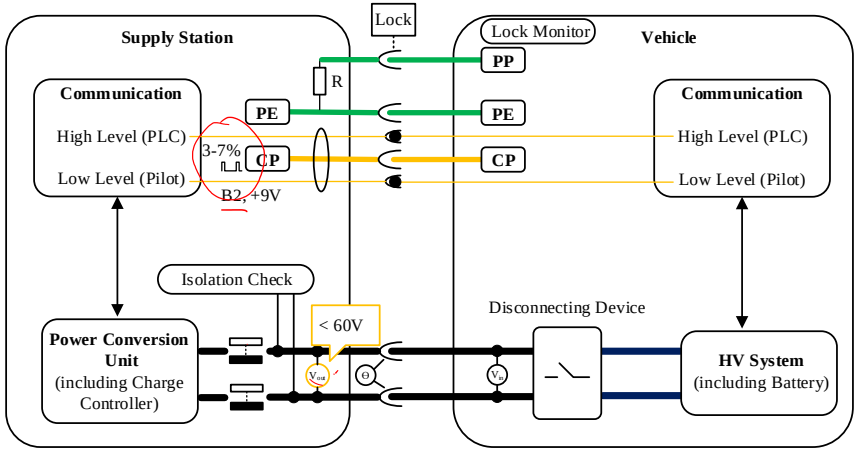
$t = t_0$

$R_{CP-PE} = 2740\ \Omega$

- Charger plug is mated at t_0
- Charging enters state B1 instantly with mating
- Vehicle is immobilized (after detection in PP)

Source: 1. https://morth.gov.in/sites/default/files/ASI/320201825106PMAIS_138_Part_2_F.pdf
2. https://www.charin.global/media/pages/technology/ccs-specification/42a9d61e04-1626949173/design_guide_combined_charging_system_v7.pdf

DC CCS2 Charging Sequence Cont.



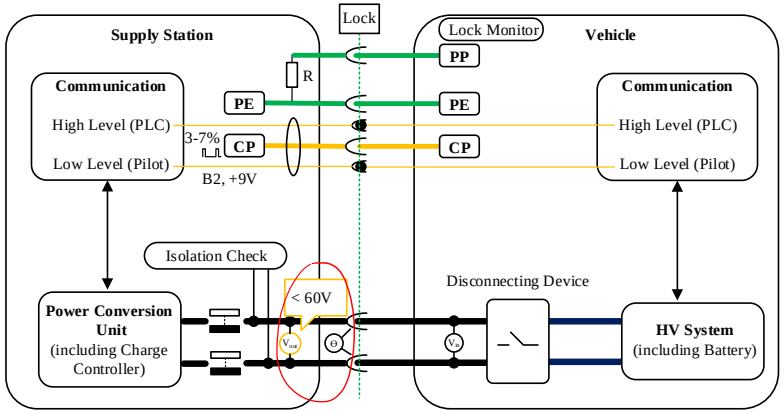
Initializing stage between $t_0 - t_1$

EV sends its maximum limits (amongst other parameters) for DC supply output current and voltage

- High level communication (PLC) starts and handshaking with exchange of charging parameter takes place.
- Shutdown if DC Voltage > 60V or incompatibility of EV and DC supply is detected

Source: 1. https://morth.gov.in/sites/default/files/ASI/320201825106PMAIS_138_Part_2_F.pdf
2. https://www.charin.global/media/pages/technology/ccs-specification/42a9d61e04-1626949173/design_guide_combined_charging_system_v7.pdf

DC CCS2 Charging Sequence Cont.



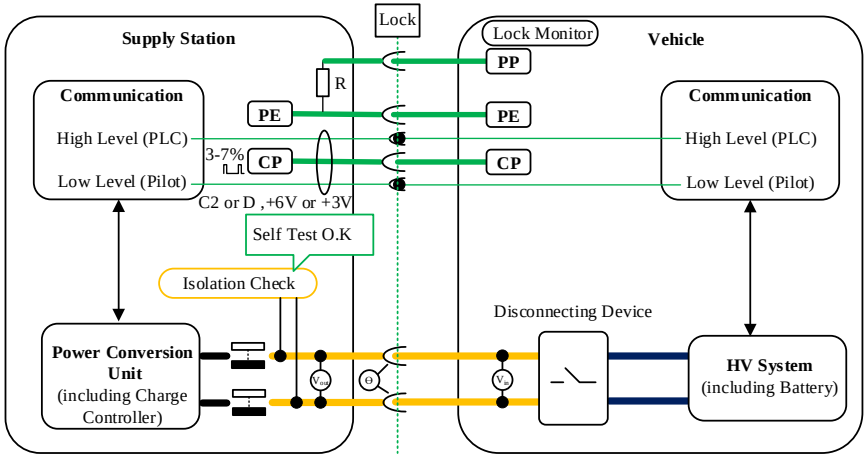
Stage between $t_1 - t_2$

- If EV and DC supply are not compatible, then the vehicle will not go to Ready, and will transition to normal shutdown sequence.

- EV locks vehicle connector in its inlet.
- Maximum values of the DC supply are responded to the EV
- DC supply can check internal insulation as long as no voltage is applied to the connector.

Source: 1. https://morth.gov.in/sites/default/files/ASI/320201825106PMAIS_138_Part_2_F.pdf
2. https://www.charin.global/media/pages/technology/ccs-specification/42a9d61e04-1626949173/design_guide_combined_charging_system_v7.pdf

DC CCS2 Charging Sequence Cont.

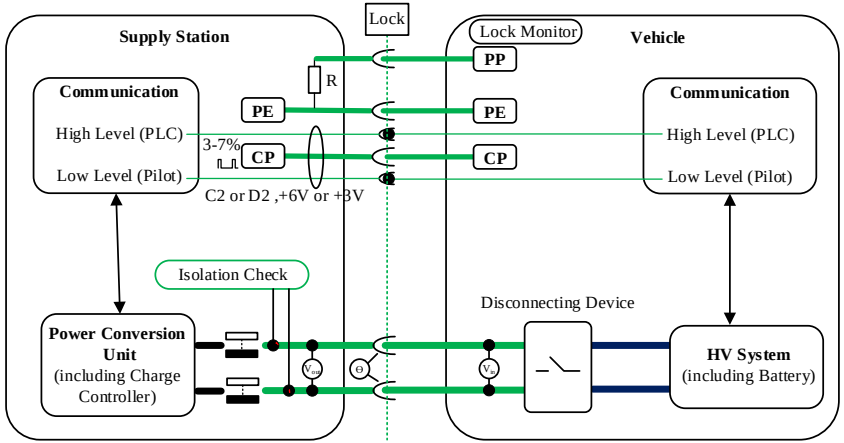


Cable check stage between $t_2 - t_3$

- EV changes CP state from B to C2/D and sets EV status "Ready"
- EV requests the cable and insulation check after connector lock has been confirmed
- DC supply starts checking HV system isolation and continuously reports isolation state

Source: 1. https://morth.gov.in/sites/default/files/ASI/320201825106PMAIS_138_Part_2_F.pdf
2. https://www.charin.global/media/pages/technology/ccs-specification/42a9d61e04-1626949173/design_guide_combined_charging_system_v7.pdf

DC CCS2 Charging Sequence Cont.

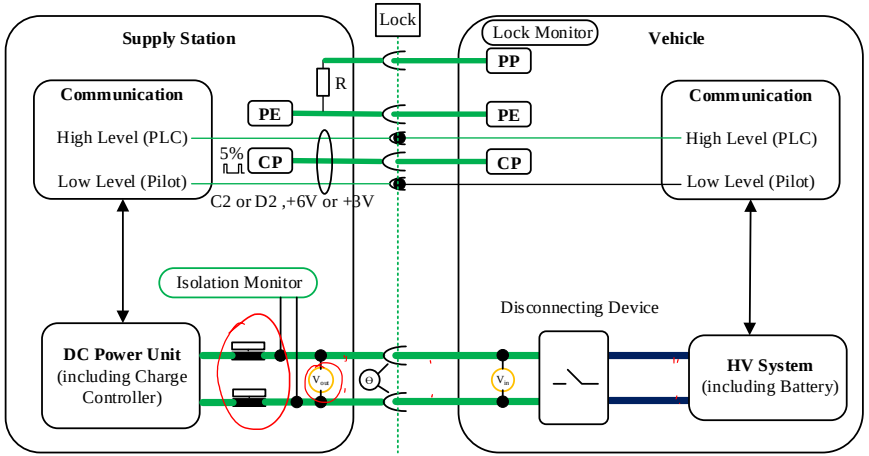


Cable check stage between t_3 - t_4

- D.C. supply determines that isolation resistance of system is above 100 kΩ.
- After successful isolation check, DC supply indicates status ”Valid” and changes status to “Ready” with Cable Check Response.

Source: 1. https://morth.gov.in/sites/default/files/ASI/320201825106PMAIS_138_Part_2_F.pdf
2. https://www.charin.global/media/pages/technology/ccs-specification/42a9d61e04-1626949173/design_guide_combined_charging_system_v7.pdf

DC CCS2 Charging Sequence Cont.



Pre-charge at t_5

Maximum current limit & required DC voltage

- Start of pre-charge stage
- EV sends Pre-Charge Request, which contains both requested DC current < 2A (maximum inrush current) and requested DC voltage.

Source: 1. https://morth.gov.in/sites/default/files/ASI/320201825106PMAIS_138_Part_2_F.pdf
2. https://www.charin.global/media/pages/technology/ccs-specification/42a9d61e04-1626949173/design_guide_combined_charging_system_v7.pdf

Thank You